

LIBRARY MANAGEMENT SYSTEM

AIM

To design and implement a Library Management System using Oracle SQL and PL/SQL to manage books, authors, categories, members, and loans.

ALGORITHM

1. Start the program.
2. Create tables for Categories, Authors, Books, Members, and Loans with appropriate constraints.
3. Create sequences to generate unique IDs.
4. Insert sample records into all tables.
5. Display books, members, available books, and book details using SQL queries.
6. Use JOIN, LIKE, COUNT, SUM, GROUP BY, and LEFT JOIN operations to retrieve useful information.
7. Display current loans, loan history, books never borrowed, and overdue books.
8. Calculate fines for overdue books using PL/SQL.
9. Create a PL/SQL block to display the library summary.
10. Use a cursor to display book titles with author names.
11. Create and display a view containing book, author, and category details.
12. Commit the changes and stop the program.

RESULT

Thus, the Library Management System was successfully implemented using Oracle SQL and PL/SQL. The system successfully performed book and member management, loan tracking, searching, fine calculation, aggregate operations, cursor processing, and view creation.